Parallel Processing Concepts

Q. What is Parallel Processing? Discuss the need for parallel processing, its types, and its primary advantages in computer architecture.

Definition to Remember

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1. Need for Parallel Processing

Historically, processors became faster by increasing the clock frequency. However, this approach hit physical limits (the power wall), causing extreme heat and power consumption.

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2. Types of Parallelism

1. Instruction-Level Parallelism (ILP):

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3. Advantages of Parallel Processing

* **Increased Speed / Throughput:** Executing tasks simultaneously drastically reduces total execution time.

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Must-Write Points (for 10 marks)

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Quick Recall

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